LECTURE 16

Random Variables

Numerical functions of random samples and their properties; sampling variability.

Presented by Dr . Devan Becker, Data200

Adapted from Data 100, Summer 2023 @ UC Berkeley by Bella Crouch and Dominic Liu

Content credit: Acknowledgments

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Today's Roadmap

Random Variables and Distributions

Expectation and Variance

Sums of Random Variables

- Example: Bernoulli and Binomial Random Variables

Populations and Samples

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Random Variables and Distributions

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Expectation and Variance

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Populations and Samples

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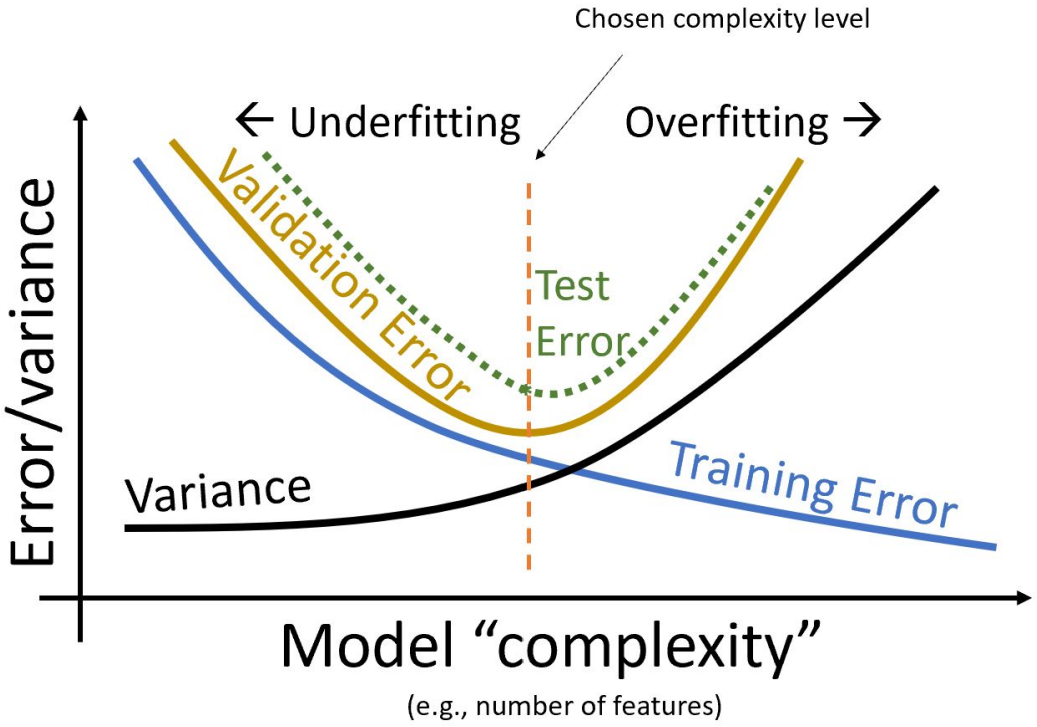
The Bias-Variance Tradeoff

What is the mathematical underpinning of this plot?



We'll come back to this…

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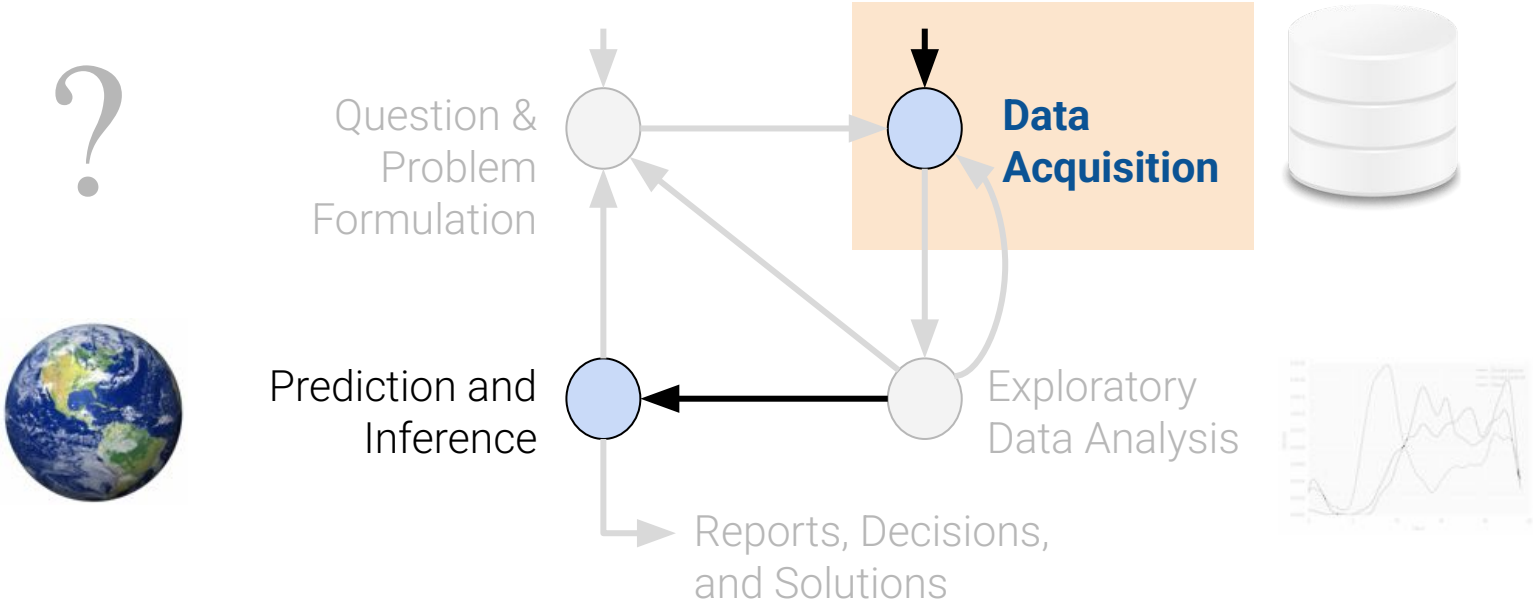
Why Probability?



(today)

Model Selection Basics:
Cross Validation
Regularization

Probability I:
Random Variables
Estimators

Probability II:
Bias and Variance
Inference/Multicollinearity

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We Will See . . .

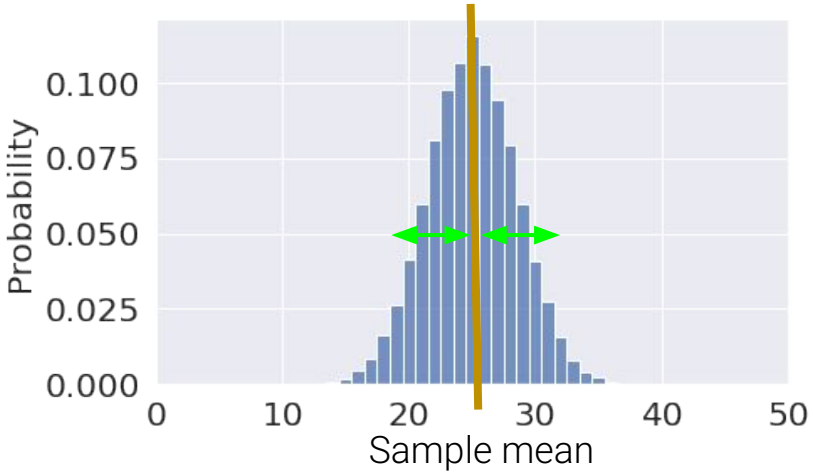
Formalize the notions of **sample statistic, population parameter** from Data 100.

From Data 8:

1. def **sample mean** - the mean of your random sample. `np.mean(data)`

2. The **Central Limit Theorem**: If you draw a large random sample with replacement, then, regardless of the population distribution, the probability distribution of the sample mean:

- Is roughly normal
- Is centered at the **population mean**
- Has an SD = $\dfrac{\text{population SD}}{\sqrt{\text{sample size}}}$



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We Will See → One Important Probability Concept

Formalize the notions of **sample statistic, population parameter** from Data 8.

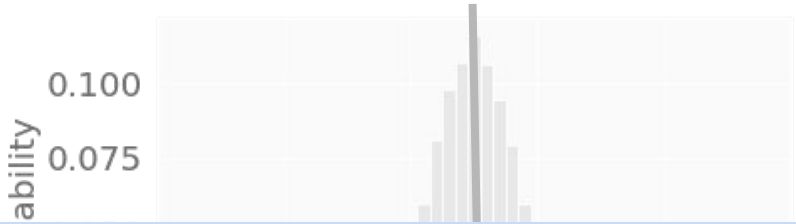
From Data 8:

1. def **sample mea**

Random Variable

2. The Central Limit Theorem: If you draw a **large random sample with replacement**, then, regardless of the population distribution, the **probability distribution** of the **sample mean**:

- Is roughly normal
- Is centered at the **population mean**
- Has an SD = $\frac{\text{population SD}}{\sqrt{\text{sample size}}}$



We will go over **just enough probability** to help you understand its implications for modeling.

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[Terminology] Random Variable

Suppose we generate random data, like a random sample from some population, then

A **random variable** is a numerical function of the randomness in the data.

- It is **random** since our sample was drawn at random.
- It is a **variable** since because its value depends on how the sample came out.

- **Domain (input)**: all possible random samples/ all possible outcomes of a random process
- **Range (output)**: number line.
- Often denoted with uppercase "variable-like" letters (e.g. X, Y).

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Example: Tossing a coin

A fair coin can land either heads (H) or tails (T), each with probability 0.5. With these possible outcomes, we can define a random variable X:

$$X = \begin{cases} 1, & \text{if coin lands heads} \\ 0, & \text{if coin lands tails} \end{cases}$$

X is a function with

- **Domain (input)**: $\{H, T\}$
- **Range (output)**: $\{1, 0\}$

We can write this in function notation: $\begin{cases} X(H) &= 1 \\ X(T) &= 0 \end{cases}$

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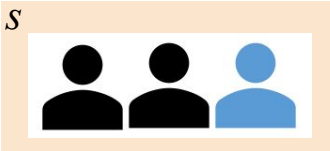
Example: Sampling Data 200 Students

Suppose we draw a simple random sample s of size 3 from the following population:

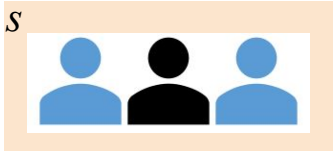
Population: all Data 200 students

Blue: data science students

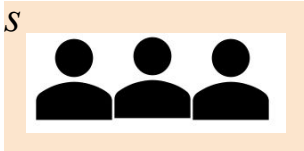
Draw several random samples:



$Y(s) = 1$



$Y(s) = 2$



$Y(s) = 0$

$Y(s) = 2$

We can define Y as the number of data science students in our sample.

- **Domain (input)**: all possible samples of size 3
- **Range (output)**: {0, 1, 2, 3}

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[Terminology] Distribution

For any random variable, we need to be able to specify two things:

- **Possible values**: the set of values the random variable can take on
- **Probabilities**: the set of probabilities describing how the total probability of 100% is split over the possible values.

Assuming (for now) that X is discrete, i.e., has a finite number of possible values:

$$P(X = x)$$

The probability that random variable X takes on the value x .

$$\sum_{\text{all } x} P(X = x) = 1$$

Probabilities must sum to 1.

We can often do this using a **probability distribution table**. In the coin toss example, the probability distribution table of X:

x	P(X = x)
0	1/2
1	1/2

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[Terminology] Distribution

The **distribution** of a random variable X is a description of how the total probability of 100% is split over all the possible values of X.

A distribution fully defines a random variable.

	Sample s	$Y(s)$
0	(1, 1, 1)	3
1	(1, 1, 0)	2
2	(1, 1, 0)	2
3	(1, 1, 0)	2
4	(1, 1, 0)	2
...	...	...
115	(0, 0, 0)	0
116	(0, 0, 0)	0
117	(0, 0, 0)	0
118	(0, 0, 0)	0
119	(0, 0, 0)	0

$$P(Y = y) = \frac{\text{number of samples where } Y = y}{\text{number of possible samples}}$$

	y	$P(Y = y)$
0	0	0.291667
1	1	0.525000
2	2	0.175000
3	3	0.008333

Probability Distribution Table of Y

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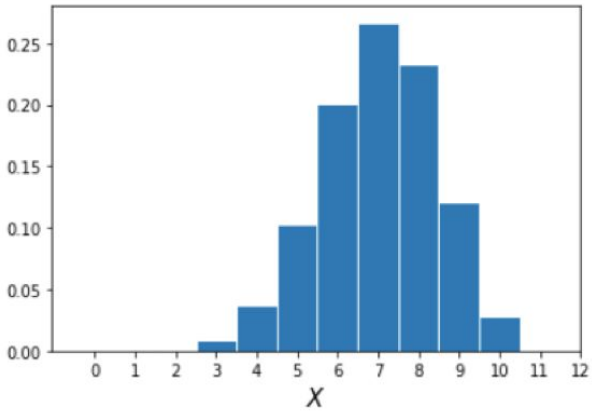
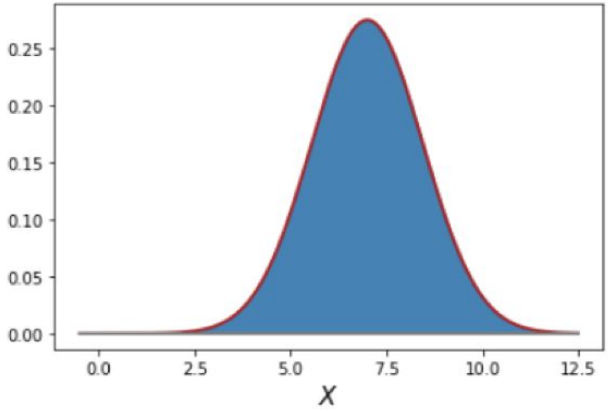
Distributions Can Be Represented as Histograms or Densities

Distribution of **discrete** random variable X



Distribution of **continuous** random variable Y



Take a probability class like Data 140 to learn more about discrete vs continuous distributions.

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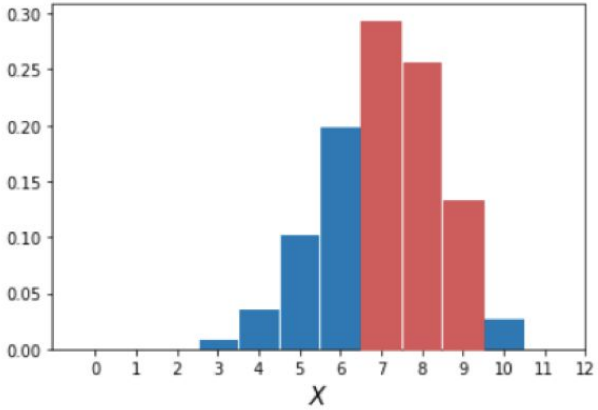
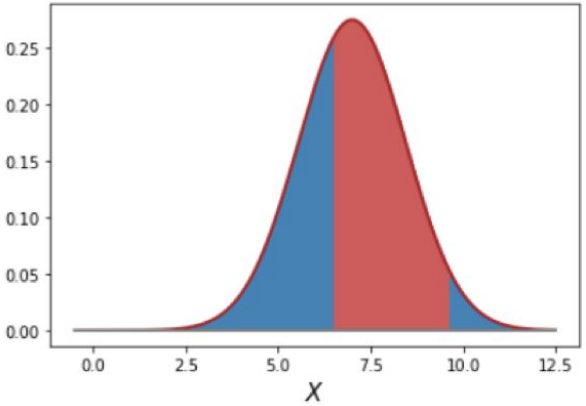
Probabilities as Areas

Distribution of **discrete** random variable X



The area of the red bars is **P(7 <= X <= 9)**.

Distribution of **continuous** random variable Y



The red area under the curve is **P(6.8 <= Y <= 9.5)**.

If we sum up the total area of the bars/under the density curve, we should get 100%, or 1.

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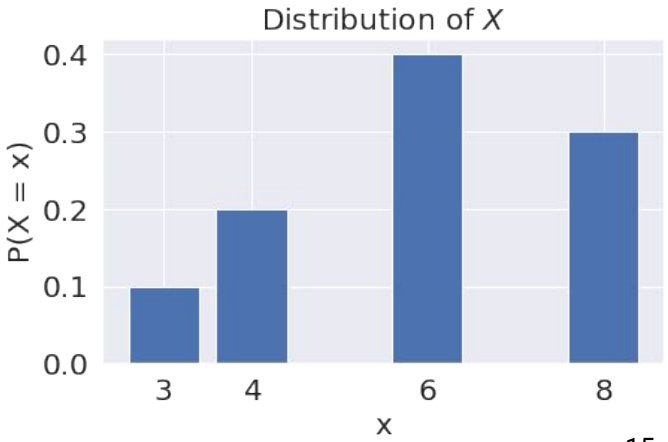
Understanding Random Variables

Compute the following probabilities for the random variable X.

x	P(X = x)
3	0.1
4	0.2
6	0.4
8	0.3

1. $P(X = 4) =$ 0.2
2. $P(X < 6) =$ $0.1 + 0.2 = 0.3$
3. $P(X \leq 6) =$ $0.1 + 0.2 + 0.4 = 0.7$
4. $P(X = 7) =$ 0
5. $P(X \leq 8) =$ 1



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Common Random Variables

Bernoulli(p)

- Takes on value 1 with probability p, and 0 with probability 1 - p.
- AKA the “indicator” random variable.

Binomial(n, p)

- Number of 1s in 'n' independent Bernoulli(p) trials.

Uniform on a finite set of values

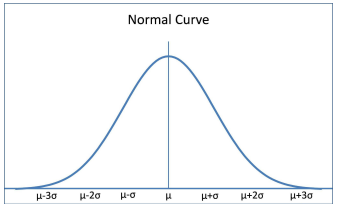
- Probability of each value is 1 / (number of possible values).
- For example, a standard/fair die.

Uniform on the unit interval (0, 1)

- Density is flat at 1 on (0, 1) and 0 elsewhere.

Normal(μ, σ^2)

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$



The numbers in parentheses are the **parameters** of a random variable, which are constants. Parameters define a random variable’s shape (i.e., distribution) and its values.

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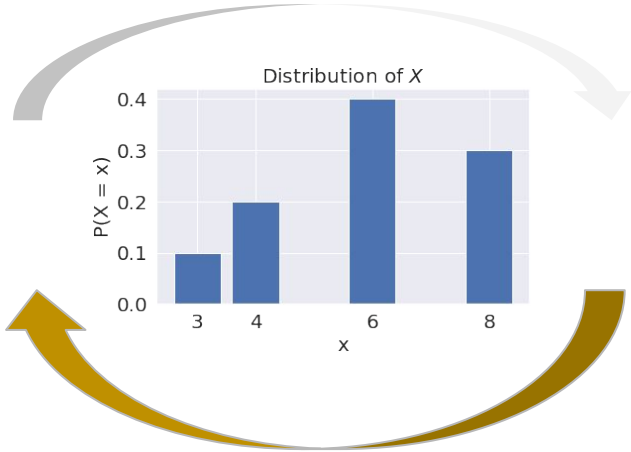
From Distribution to (Simulated) Population

?

X(s) from all possible samples?

$$P(X = x) = \frac{\#\text{ times where } X = x}{\text{pop. size}}$$



Given a random variable's distribution, how could we **generate/simulate** a population?

x	P(X = x)
3	0.1
4	0.2
6	0.4
8	0.3

Probability Distribution Table

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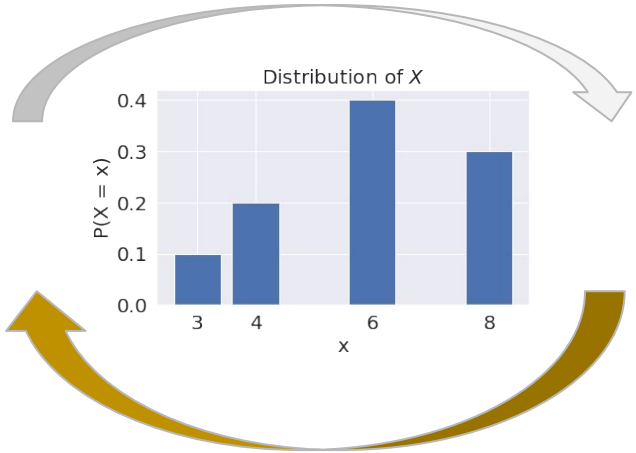
From Distribution to (Simulated) Population

	X(s)
0	3
1	4
2	4
3	6
4	8
...	...
79995	6
79996	6
79997	4
79998	6
...	...

X(s) from many, many **(simulated)** samples

$$P(X = x) = \frac{\#\text{ times where } X = x}{\text{pop. size}}$$



Simulate: Randomly pick values of X according to its distribution

`np.random.choice` or `df.sample`

x	P(X = x)
3	0.1
4	0.2
6	0.4
8	0.3

Probability Distribution Table

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Expectation and Variance

Random Variables and Distributions

Expectation and Variance

Sums of Random Variables

- Example: Bernoulli and Binomial Random Variables

Populations and Samples

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Descriptive Properties of Random Variables

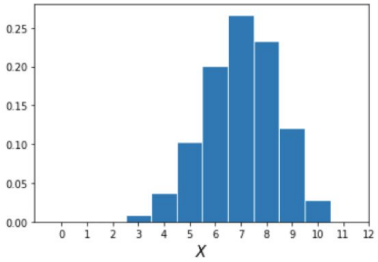
There are several ways to describe a random variable:

Table of all samples s , $X(s)$

Distribution Table

$P(X = x)$

Histogram



Definition (fully describes)

Random Variable X

Properties (summarizes)

Expectation

the "average value" of X

$$E[X]$$

Variance

the "spread" of X

$$\mathrm{Var}(X)$$

The expectation and variance of a random variable are **numerical summaries** of X.

They are numbers and are not random!

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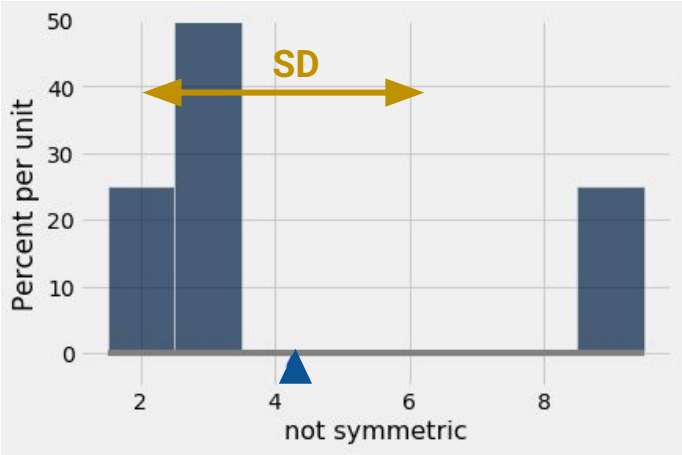
You've Seen This Before

> The **mean** (Data 200: **expectation**) is the center of gravity or **balance point** of the **histogram** (Data 200: of a random variable). [textbook]

> The **variance** is a measure of spread.
> It is the **expected squared deviation from the mean** (Data 200: of a random variable). [textbook]



In Data 100, you computed these from the datapoints themselves (i.e., the sample of data).

In Data 200, we redefine these terms with respect to probability distributions.

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Definition of Expectation

The **expectation** of a random variable X is the **weighted average** of the possible values of X, where the weights are the probabilities of the values.

Two equivalent ways to apply the weights:

1. One sample at a time:

$$E[X] = \sum_{\text{all samples } s} X(s)P(s)$$

2. One possible value at a time:

$$E[X] = \sum_{\text{all possible } x} xP(X = x)$$

More common (we are usually given the distribution, not all possible samples)

Expectation is a number, not a random variable!

- It is analogous to the **average** (same units as the random variable).
- It is the center of gravity of the probability distribution histogram.
- If we simulate the variable many times, it is the long run average of the random variable

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Example 1: Tossing a coin

A fair coin can land either heads (H) or tails (T), each with probability 0.5. With these possible outcomes, we can define a random variable X:

$$X = \begin{cases} 1, & \text{if coin lands heads} \\ 0, & \text{if coin lands tails} \end{cases}$$

$$\begin{aligned} E[X] &= \sum_x x P(X = x) \\ &= 1 \cdot 0.5 + 0 \cdot 0.5 \\ &= 0.5 \end{aligned}$$

Note, $E[X] = 0.5$ is not a possible value of X! It is an average.

The expectation of X **does not** need to be a possible value of X.

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Example 2

Consider the random variable X we defined earlier.

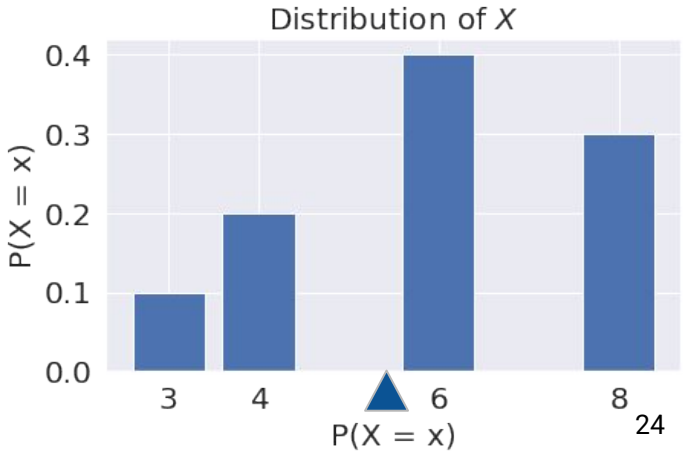
x	P(X = x)
3	0.1
4	0.2
6	0.4
8	0.3

$$E[X] = \sum_x xP(X = x)$$

$$= 3 \cdot 0.1 + 4 \cdot 0.2 + 6 \cdot 0.4 + 8 \cdot 0.3$$

$$= 0.3 + 0.8 + 2.4 + 2.4$$

$$= 5.9$$



Note, $E[X] = 5.9$ is not a possible value of X!
It is an average.

The expectation of X does not need to be a possible value of X.

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Definition of Variance

Variance is the **expected squared deviation from the expectation** of X.

$$\mathrm{Var}(X) = E\left[(X - E[X])^2\right]$$

- The units of the variance are the square of the units of X.
- To get back to the right scale, use the **standard deviation** of X: $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$

Variance is a number, not a random variable!

- The main use of variance is to **quantify chance error**. How far away from the expectation could X be, just by chance?

By Chebyshev's inequality:

- No matter what the shape of the distribution of X is, the vast majority of the probability lies in the interval "expectation plus or minus a few SDs."

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Variance: Alternate Calculation

$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$

There's a more convenient form of variance:

$$\mathrm{Var}(X) = E\left[X^2\right] - (E[X])^2$$

- Called the "computation formula" for variance
- Proof (involves expanding the square and properties of expectation/summations): link
- Useful in Mean Squared Error calculations
  - If X is centered (i.e. E[X] = 0), then $E[X^2] = \mathrm{Var}(X)$
- When computing variance by hand, often used instead of definition.

How do we calculate $E\left[X^2\right]$?

X^2 is just another random variable, with its own probability distribution. We can calculate its expectation as

$$E\left[X^2\right] = \sum_x x^2 P(X = x)$$

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Example: Dice

Let X be the outcome of a single dice roll.

X is a random variable.

$$P(X = x) = \begin{cases} 1/6 & \text{if } x \in \{1, 2, 3, 4, 5, 6\} \\ 0 & \text{otherwise} \end{cases}$$

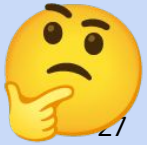
1. What is the expectation, $E[X]$?

$$\mathbb{E}[X] = \sum_x x P(X = x)$$

$$\begin{aligned} \mathrm{Var}(X) &= \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

(definitions/properties)

2. What is the variance, $\mathrm{Var}(X)$?



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Example: Dice Is the Plural; Dice Is the Singular

Let X be the outcome of a single dice roll.

X is a random variable.

$$P(X = x) = \begin{cases} 1/6 & \text{if } x \in \{1, 2, 3, 4, 5, 6\} \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{E}[X] = \sum_x x P(X = x)$$

$$\begin{aligned} \mathrm{Var}(X) &= \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

1. What is the expectation, $E[X]$?

$$\begin{aligned} \mathbb{E}[X] &= 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6) \\ &= (1/6)(1 + 2 + 3 + 4 + 5 + 6) = \frac{7}{2} \end{aligned}$$

2. What is the variance, $\mathrm{Var}(X)$?

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Example: Dice Is the Plural; Dice Is the Singular

Let X be the outcome of a single dice roll.

X is a random variable.

$$P(X = x) = \begin{cases} 1/6 & \text{if } x \in \{1, 2, 3, 4, 5, 6\} \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbb{E}[X] = \sum_x x P(X = x)$$

$$\begin{aligned} \mathrm{Var}(X) &= \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

1. What is the expectation, $E[X]$?

$$\begin{aligned} \mathbb{E}[X] &= 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6) \\ &= (1/6)(1 + 2 + 3 + 4 + 5 + 6) = \frac{7}{2} \end{aligned}$$

2. What is the variance, $\mathrm{Var}(X)$?

Approach 1: Definition

$$\begin{aligned} \mathrm{Var}(X) = (1/6)\big((1 - 7/2)^2 + (2 - 7/2)^2 \\ + (3 - 7/2)^2 + (4 - 7/2)^2 \\ + (5 - 7/2)^2 + (6 - 7/2)^2\big) \\ = 35/12 \end{aligned}$$

Approach 2: Property

$$\mathbb{E}[X^2] = \sum_x x^2 P(X = x) \quad = 91/6$$

$$\mathrm{Var}(X) = 91/6 - (7/2)^2 = 35/12$$

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Sums of Random Variables

Random Variables and Distributions

Expectation and Variance

Sums of Random Variables

- Example: Bernoulli and Binomial Random Variables

Populations and Samples

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Functions of Multiple Random Variables

A function of a random variable is also a random variable!

If you create multiple random variables based on your sample…

…then functions of those random variables are also random variables.

For instance, if $X_1, X_2, \ldots, X_n$ are random variables, then so are all of these:

$$X_n^2 \qquad \#\{i : X_i > 10\}$$

$$\max(X_1, X_2, \ldots, X_n) \qquad \frac{1}{n}\sum_{i=1}^{n}(X_i - c)^2$$

$$\boxed{\frac{1}{n}\sum_{i=1}^{n} X_i}$$

Many functions of RVs that we care about (**counts**, **means**) involve **sums of RVs**, so we expand on properties of sums of RVs.

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Equal vs. Identically Distributed vs. i.i.d.

Suppose that we have two random variables X and Y.

X and Y are **equal** if:

- $X(s) = Y(s)$ for every sample s.
- We write $X = Y$.

X and Y are **identically distributed** if:

- The distribution of X is the same as the distribution of Y
- We say "X and Y are equal in distribution."
- If $X = Y$, then X and Y are identically distributed;
  but the converse is not true (ex: **Y = 7-X**, X is a die)

X and Y are **independent and identically distributed (i.i.d.)** if:

- X and Y are identically distributed, *and*
- Knowing the outcome of X does not influence your belief
  of the outcome of Y, and vice versa ("X and Y are independent.")
- Independence is covered more in statistics courses.
- In Data 200, you will never be expected to prove that RVs are i.i.d.

i.i.d. RVs

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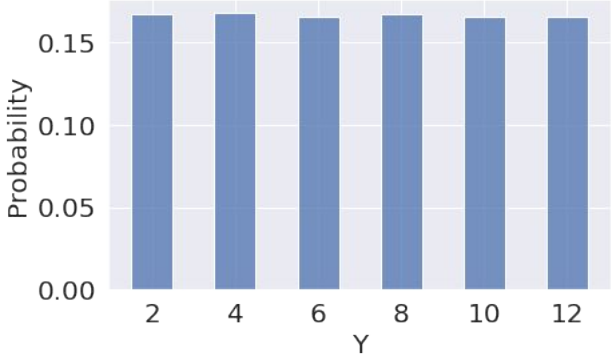
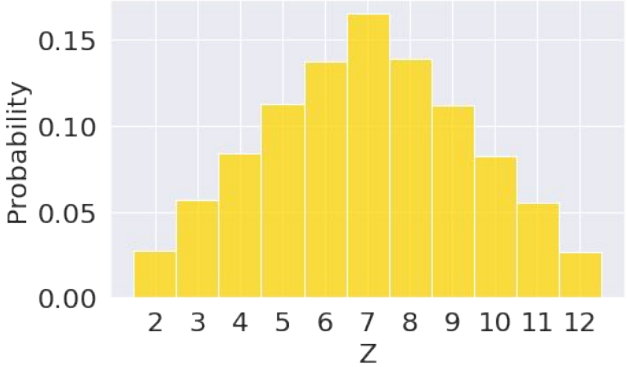
Distributions of Sums

Let X_1 and X_2 be numbers on rolls of two dice.

- X_1, X_2 are **i.i.d.**, so X_1, X_2 have the same distribution.
- But the sums $Y = X_1 + X_1 = 2X_1$ and $Z = X_1 + X_2$ have different distributions!

Let's show this through simulation:





- Same expectation…
- But $Y = 2X_1$ has larger variance!

	Y	Z
E[•]	6.984400	6.984950
Var(•)	11.669203	5.817246
SD(•)	3.416021	2.411897

How can we directly compute $E[Y]$, $\text{Var}(Y)$, **without** simulating distributions?

Demo

CC BY NC SA

Properties of Expectation [1/3]

Instead of simulating full distributions, we often just compute expectation and variance directly.

Recall definitions of expectation:

$$E[X] = \sum_{\text{all possible } x} xP(X = x)$$

Properties:

1. **Expectation is linear**.
Intuition: summations are linear. Proof

$$E[aX + b] = aE[X] + b$$

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Properties of Expectation [2/3]

Instead of simulating full distributions, we often just compute expectation and variance directly.

Recall definitions of expectation:

$$E[X] = \sum_{\text{all possible } x} xP(X = x)$$

Properties:

1. **Expectation is linear**.
Intuition: summations are linear. Proof

$$E[aX + b] = aE[X] + b$$

2. Expectation is linear in sums of RVs, for any relationship between X and Y. Proof

$$E[X + Y] = E[X] + E[Y]$$

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Properties of Expectation [3/3]

Instead of simulating full distributions, we often just compute expectation and variance directly.

Recall definitions of expectation:

$$E[X] = \sum_{\text{all possible } x} x P(X = x)$$

Properties:

1. **Expectation is linear**.
Intuition: summations are linear. Proof

$$E[aX + b] = aE[X] + b$$

2. Expectation is linear in sums of RVs,
for any relationship between X and Y. Proof

$$E[X + Y] = E[X] + E[Y]$$

3. If g is a non-linear function, then in general $E[g(X)] \neq g(E[X])$

- Example: if X is -1 or 1 with equal probability, then $E[X] = 0$ but $E[X^2] = 1 \neq 0$.

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Properties of Variance [1/2]

Recall definition of variance: $\mathrm{Var}(X) = E\left[(X - E[X])^2\right]$

Properties:

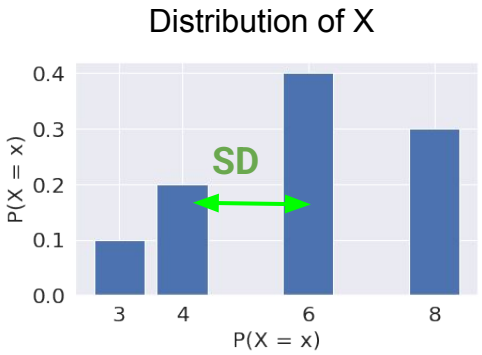
1. **Variance is non-linear**:

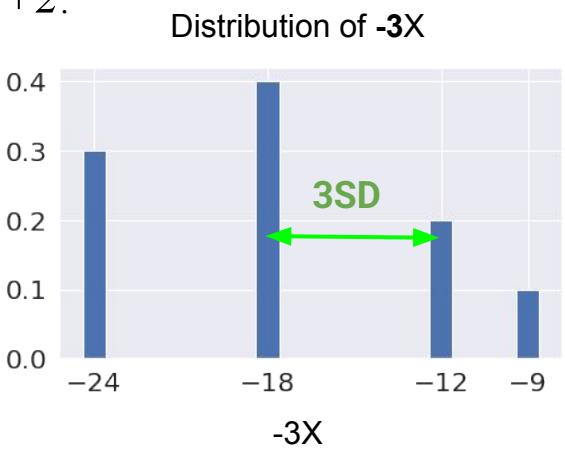
Shifting doesn't change variance, scaling does

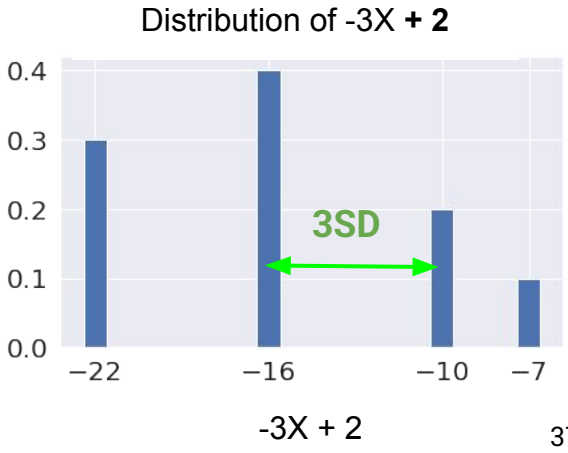
$$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

$$\mathrm{SD}(aX + b) = |a|\mathrm{SD}(X)$$

Intuition (full proof): Consider the Standard Deviation for $Y = -3X+2$:







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Properties of Variance [2/2]

Recall definition of variance: $$\mathrm{Var}(X) = E\left[(X - E[X])^2\right]$$

Properties:

1. **Variance is non-linear**:

$$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

Intuition (full proof): Consider the Standard Deviation for $Y = -3X+2$:

$$\mathrm{SD}(aX + b) = |a|\mathrm{SD}(X)$$

2. Variance of sums of RVs is affected by the (in)dependence of the RVs (derivation):

$$\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 2\underbrace{\mathrm{Cov}(X, Y)}$$

Covariance of X and Y (next slide).
If X, Y independent,
then Cov(X, Y) = 0.

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Covariance and Correlation: The Basics

Covariance is the expected product of deviations from expectation.

$$\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

- A generalization of variance. Note $\mathrm{Cov}(X, X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathrm{Var}(X)$.
- Interpret by defining **correlation** (yes, *that* correlation!):

$$r(X, Y) = \mathbb{E}\left[\underbrace{\left(\frac{X - \mathbb{E}[X]}{\mathrm{SD}(X)}\right)}_{X \text{ in } \textbf{standard units} \text{ (link)}}\left(\frac{Y - \mathbb{E}[Y]}{\mathrm{SD}(Y)}\right)\right] = \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\mathrm{SD}(Y)}$$

Correlation (and therefore covariance) measures a linear relationship between X and Y.

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Covariance and Correlation: The Basics

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- Interpret by defining **correlation** (yes, *that* correlation!):

$$r(X, Y) = \mathbb{E}\left[\underbrace{\left(\frac{X - \mathbb{E}[X]}{\mathrm{SD}(X)}\right)}_{\text{standard units of } X \text{ (link)}}\left(\frac{Y - \mathbb{E}[Y]}{\mathrm{SD}(Y)}\right)\right] = \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\mathrm{SD}(Y)}$$

Correlation (and therefore covariance) measures a linear relationship between X and Y.

- If X and Y are correlated, then knowing X tells you something about Y.
- "X and Y are uncorrelated" is the same as "Correlation and covariance equal to 0"
- **Independent X, Y are uncorrelated**, because knowing X tells you nothing about Y.
- The converse is not necessarily true: **X, Y could be uncorrelated but not independent**.

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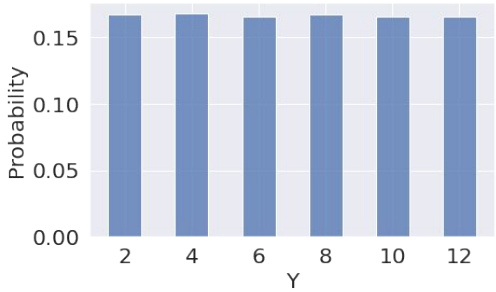
Dice, Our Old Friends: Expectation

Let X_1 and X_2 be numbers on one roll of two dice.

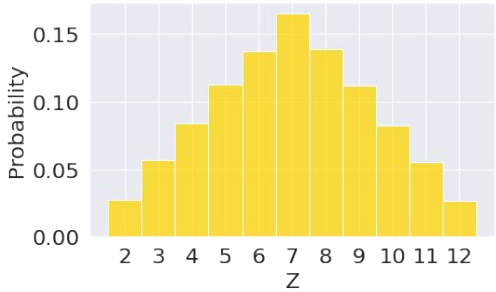
- X_1, X_2 are **i.i.d.**, so X_1, X_2 have the same distribution.
- Therefore $E[X_1] = E[X_2] = 7/2 \quad \mathrm{Var}(X_1) = \mathrm{Var}(X_2) = 35/12$

$$\boxed{Y = 2X_1}$$



$$E[Y] = E[2X_1] = 2E[X_1] = 7$$

$$\boxed{Z = X_1 + X_2}$$



$$E[Z] = E[X_1 + X_2] = E[X_1] + E[X_2] = 7$$

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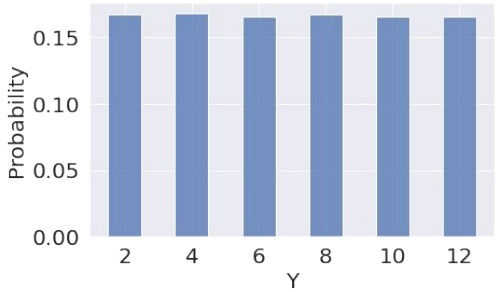
Dice, Our Old Friends: Variance

Let X_1 and X_2 be numbers on two rolls of a die.

- X_1, X_2 are **i.i.d.**, so X_1, X_2 have the same distribution.
- Therefore $E[X_1] = E[X_2] = 7/2$ $\quad \text{Var}(X_1) = \text{Var}(X_2) = 35/12$

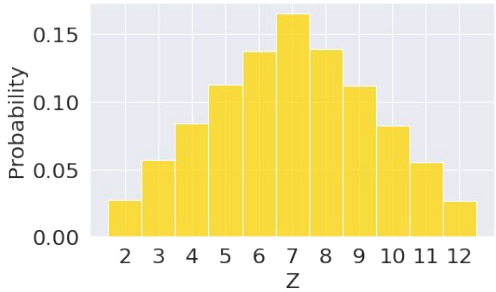
$$\boxed{Y = 2X_1}$$



$$E[Y] = E[2X_1] = 2E[X_1] = 7$$

$$\text{Var}(Y) = \text{Var}(2X_1) = 4\text{Var}(X_1)$$
$$= 4(35/12)$$
$$\approx \mathbf{11.67}$$

$$\boxed{Z = X_1 + X_2}$$



$$E[Z] = E[X_1 + X_2] = E[X_1] + E[X_2] = 7$$

$$\text{Var}(Z) = \text{Var}(X_1) + \text{Var}(X_2) + 2\cancel{\text{Cov}(X1, X2)} \rightarrow 0$$
$$= (35/12) + (35/12) + 0$$
$$\approx \mathbf{5.83}$$

X_1, X_2 independent

Simulation Results

	Y	Z
E[•]	6.984400	6.984950
Var(•)	11.669203	5.817246
SD(•)	3.416021	2.411897

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[Summary] Expectation and Variance for Linear Functions of Random Variables

Let X be a random variable with distribution $P(X = x)$.

$$\mathbb{E}[X] = \sum_x x P(X = x)$$

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \quad \text{(definition)}$$

$$= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \quad \text{(easier computation)}$$

Let a and b be scalar values.

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

$$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

Let Y be another random variable.

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

$$\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \underbrace{2\mathrm{Cov}(X, Y)}_{\text{0 if X, Y independent.}}$$

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Interlude

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Bernoulli and Binomial Random Variables

Random Variables and Distributions

Expectation and Variance

Sums of Random Variables

- **Example: Bernoulli and Binomial Random Variables**

Populations and Samples

BY NC SA

Common Random Variables

Bernoulli(p)

- Takes on value 1 with probability p, and 0 with probability 1 - p
- AKA the “indicator” random variable.

Binomial(n, p)

- Number of 1s in n independent Bernoulli(p) trials

Uniform on a finite set of values

- Probability of each value is 1 / (size of set)
- For example, a standard die

Uniform on the unit interval(0, 1)

- Density is flat on (0, 1) and 0 elsewhere

Normal(μ, σ^2)

We’ll now revisit these to solidify our understanding of expectation/variance.

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Properties of Bernoulli Random Variables

$$\mathbb{E}[X] = \sum_x xP(X = x)$$
$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$$
$$= \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Definitions

Let X be a **Bernoulli**(p) random variable.

- Takes on value 1 with probability p, and 0 with probability 1 - p.
- AKA the "indicator" random variable.

Expectation:

$$\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p$$

We will get an average value of p across many, many samples

Variance:

$$\mathbb{E}[X^2] = 1^2 \cdot p + 0 \cdot (1 - p) = p$$
$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$
$$= p - p^2 = p(1 - p)$$

Lower Var: p = 0.1 or 0.9
Higher Var: p close to 0.5

More info: google("plot x(1 - x)")

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Properties of Binomial Random Variables

Let Y be a **Binomial**(n, p) random variable.

- Y is the number (i.e., count) of 1s in n independent Bernoulli(p) trials.
- Distribution of Y given by the binomial formula.

We can write: $$Y = \sum_{i=1}^{n} X_i$$

A count is a **sum** of 0's and 1's.

- X_i is the indicator of success on trial i. X_i = 1 if trial i is a success, else 0.
- All X_i s are **i.i.d.** (independent and identically distributed) and **Bernoulli**(p).

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Properties of Binomial Random Variables

Let Y be a **Binomial**(n, p) random variable.

- Y is the number (i.e., count) of 1s in n independent Bernoulli(p) trials.
- Distribution of Y given by the binomial formula (Lecture 2).

We can write: $$Y = \sum_{i=1}^{n} X_i$$

A count is a sum of 0's and 1's.

- X_i is the indicator of success on trial i. X_i = 1 if trial i is a success, else 0.
- All X_i s are **IID** (independent and identically distributed) and **Bernoulli**(p).

Expectation: $$\mathbb{E}[Y] = \sum_{i=1}^{n} \mathbb{E}[X_i] = np$$

Variance: Because all X_i s are independent, $\mathrm{Cov}(X_i, X_j) = 0$ for all i, j.

$$\mathrm{Var}(Y) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = np(1-p)$$

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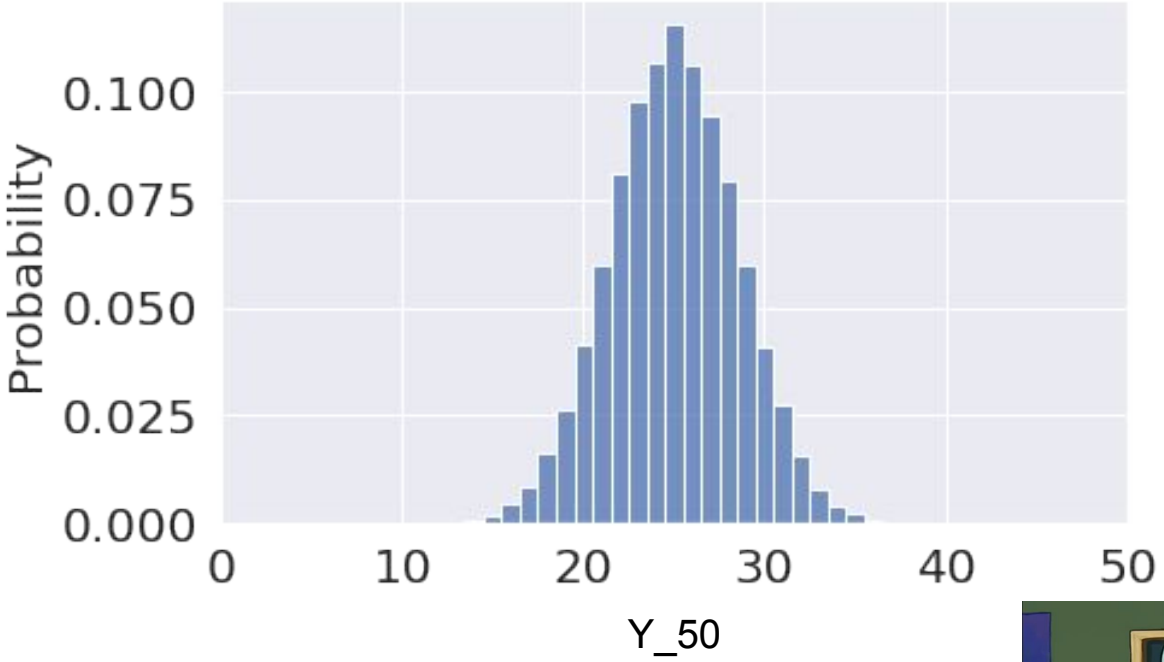
Binomial(n, p) for Large n

For p = 0.5, n = 50 (i.e. number of heads in 50 fair coin flips):



Hallgrímskirkja in Iceland

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Populations and Samples

Random Variables and Distributions

Expectation and Variance

Sums of Random Variables

- Example: Bernoulli and Binomial Random Variables

Populations and Samples

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From Populations to Samples

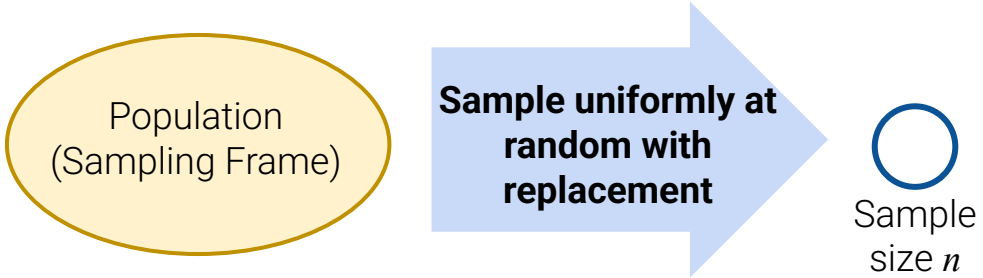
Today, we've talked extensively about **populations**:

- If we know the **distribution of a random variable**, we can reliably compute expectation, variance, functions of the random variable, etc.

However, in Data Science, we often collect **samples**.

- We don't know the distribution of our population.
- We'd like to use the distribution of your sample to estimate/infer properties of the population.

The **big assumption** we make in modeling/inference:



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The Sample is a Set of i.i.d. Random Variables

Population (Sampling Frame)

Sample uniformly at random with replacement

Sample size n

x	P(X = x)
3	0.1
4	0.2
6	0.4
8	0.3

or

	X(s)
0	3
1	4
2	4
3	6
4	8
...	...
79995	6
79996	6
79997	4
79998	6
79999	6
...	...

Population
(really large N)

df.sample(n, replace=True)
[documentation]

	X
0	6
1	8
2	6
3	6
4	3
...	...
95	8
96	6
97	6
98	3
99	8

Each observation in our sample is a **Random Variable** drawn **i.i.d.** from our population distribution.

Sample
(n << N)

$X_1, X_2, \ldots, X_n$

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The Sample is a Set of i.i.d. Random Variables

Population (Sampling Frame)

Sample uniformly at random with replacement

Sample size n

x	P(X = x)
3	0.1
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or

	X(s)
0	3
1	4
2	4
3	6
4	8
...	...
79995	6
79996	6
79997	4
79998	6
79999	6
...	...

$E[X] = 5.9$

Population Mean
A **number**,
i.e., fixed value

`df.sample(n, replace=True)`
[documentation]

	X
0	6
1	8
2	6
3	6
4	3
...	...
95	8
96	6
97	6
98	3
99	8

Sample Mean
A **random variable**!
Depends on our randomly drawn sample!!

`np.mean(...) = 5.71`

Sample $X_1, X_2, \ldots, X_n$

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[Terminology] Sample Mean

Consider an i.i.d. sample $X_1, X_2, \ldots, X_n$ drawn from a population with mean μ and SD σ.

Define the **sample mean**:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Expectation:

$$\mathbb{E}[\bar{X}_n] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i]$$
$$= \frac{1}{n}(n\mu) = \mu$$

Variance/Standard Deviation:

$$\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\underbrace{\left(\sum_{i=1}^{n}\mathrm{Var}(X_i)\right)}_{\text{IID} \rightarrow \mathrm{Cov}(X_i, X_j) = 0}$$
$$= \frac{1}{n^2}\left(n\sigma^2\right) = \frac{\sigma^2}{n}$$

$$\mathrm{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

Distribution?

$\bar{X}_n$ is **normally distributed** by the **Central Limit Theorem**.

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Central Limit Theorem

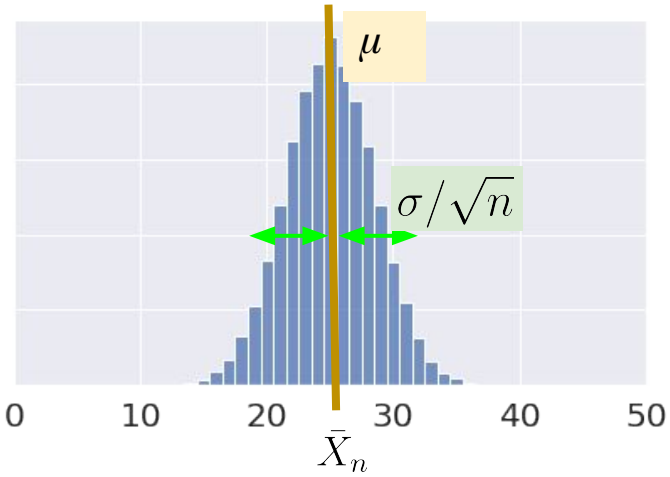
No matter what population you are drawing from:

If an i.i.d. sample of size n is large,

the probability distribution of the **sample mean**

is **roughly normal** with mean μ and SD $\sigma/\sqrt{n}$.

(Out of scope) (previous slide)



Any theorem that provides the rough distribution of a statistic
and **doesn't need the distribution of the population** is valuable to data scientists.

- Because we rarely know a lot about the population!

For a more in-depth demo: https://onlinestatbook.com/stat_sim/sampling_dist/

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How Large Is "Large"?

No matter what population you are drawing from:

If an i.i.d. **sample of size n is large**,

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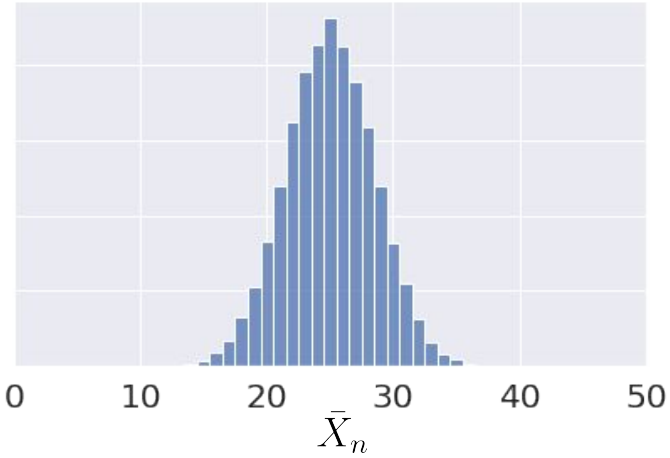

How large does n have to be for the normal approximation to be good?

- …It depends on the shape of the distribution of the population…
- If population is **roughly symmetric and unimodal**/uniform, could need as few as **n = 20**.
- If population is very skewed, you will need bigger n.
- If in doubt, you can bootstrap the sample mean and see if the bootstrapped distribution is bell-shaped.

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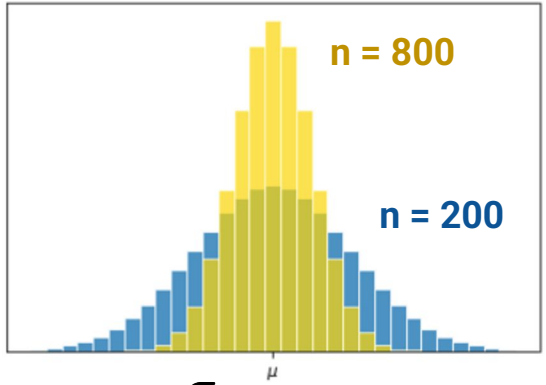
Using the Sample Mean to Estimate the Population Mean

Our goal is often to **estimate** some characteristic of a population.

- Example: average height of Cal undergraduates.
- We typically can collect a **single sample**. It has just one average.
- Since that sample was random, it *could have* come out differently.
- CLT helps us understand how this can come out differently.



We should consider the **average value and spread** of all possible sample means, and what this means for how big n should be.

$$\mathbb{E}[\bar{X}_n] = \mu$$

$$\text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

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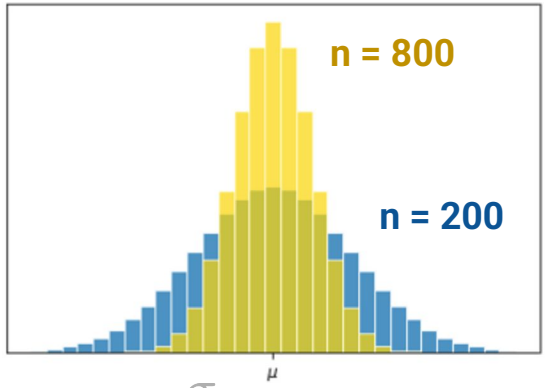

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$$\text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$$

For every sample size, the expected value of the sample mean is the population mean.

We call the sample mean an **unbiased estimator** of the population mean. (more in next lecture)

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Using the Sample Mean to Estimate the Population Mean

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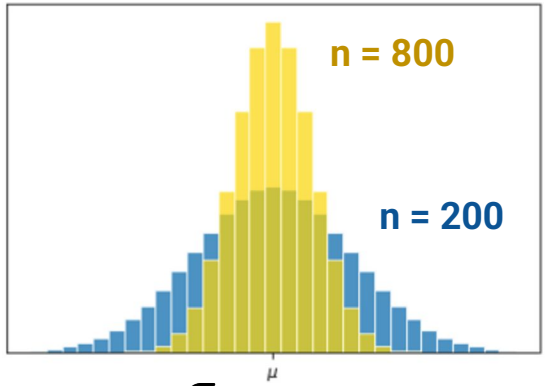

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For every sample size, the expected value of the sample mean is the population mean.

We call the sample mean an **unbiased estimator** of the population mean. (more in next lecture)

Square root law (Data 8): If you increase the sample size by a factor, the SD decreases by the square root of the factor.

The sample mean is more likely to be close to the population mean if we have a larger sample size.

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Have a Normal Day!

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[Extra Slides] Derivations

Random Variables and Distributions

- Expectation and Variance
- Equality vs Identically Distributed
- Common RVs: Bernoulli, Binomial

Functions of Random Variables

- Distributions through Simulation, I.I.D.
- Properties of Expectation and Variance
- Covariance, Correlation
- Standard Units

Sample Statistics

- Sample Mean
- Central Limit Theorem

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Standardization of random variables

Jump back: link

X in **standard units** is the random variable $$X_{su} = \frac{X - \mathbb{E}(X)}{\mathbb{SD}(X)}.$$

X_{su} measures X on the scale **"number of SDs from expectation**."

- It is a linear transformation of X. By the linear transformation rules for expectation and variance:

$$\mathbb{E}(X_{su}) = 0, \quad \mathbb{SD}(X_{su}) = 1$$

- Since X_{su} is centered (has expectation 0):

$$\mathbb{E}(X_{su}^2) = \mathbb{V}ar(X_{su}) = 1$$

You should prove these facts yourself.

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Variance: An Alternate Derivation

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There's a more convenient form of variance for use in calculations.

$$\mathbb{V}ar(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$$

To derive this, we make repeated use of the linearity of expectation.

$$\begin{aligned} Var(X) &= E((X - E(X))^2) \\ &= E\big(X^2 - 2XE(X) + (E(X)^2)\big) \\ &= E(X^2) - 2E(X)E(X) + (E(X))^2 \\ &= E(X^2) - (E(X))^2 \end{aligned}$$

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Properties of Expectation #1

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Recall definition of expectation:

$$\mathbb{E}[X] = \sum_x xP(X = x)$$

$$\mathbb{E}[X] = \sum_{\substack{\text{all samples} \\ s}} X(s)P(s)$$

1. **Expectation is linear**:

(intuition: summations are linear)

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

Proof:

$$\begin{aligned}\mathbb{E}[aX + b] &= \sum_x (ax + b)P(X = x) = \sum_x \big(axP(X = x) + bP(X = x)\big) \\ &= a\sum_x xP(X = x) + b\sum_x P(X = x) \\ &= a\mathbb{E}[X] + b \cdot 1\end{aligned}$$

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Properties of Expectation #2

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Recall definitions of expectation:

$$\mathbb{E}[X] = \sum_x x P(X = x)$$

$$\mathbb{E}[X] = \sum_{\substack{\text{all samples} \\ s}} X(s)P(s)$$

3. **Expectation is linear in sums of RVs**:

For any relationship between X and Y.

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$$

Proof:

$$\mathbb{E}[X+Y] = \sum_s (X+Y)(s)P(s) = \sum_s \big(X(s) + Y(s)\big)P(s)$$

$$= \sum_s \big(X(s)P(s) + Y(s)P(s)\big)$$

$$= \sum_s X(s)\,P(s) + \sum_s Y(s)P(s)$$

$$= \mathbb{E}[X] + \mathbb{E}[Y]$$

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Properties of Variance #1

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We know that $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$

In order to compute Var(aX + b), consider:

- A shift by **b** units **does not** affect spread. Thus, Var(aX + b) = Var(aX).
- The multiplication by **a does** affect spread!

Then,

$$\begin{aligned} Var(aX+b) = Var(aX) &= E((aX)^2) - (E(aX))^2 \\ &= E(a^2X^2) - (aE(X))^2 \\ &= a^2\left(E(X^2) - (E(X))^2\right) \\ &= a^2 Var(X) \end{aligned}$$

In summary:

$$\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$$

$$\mathbb{V}ar(aX + b) = a^2\mathbb{V}ar(X)$$
$$\mathbb{SD}(aX + b) = |a|\mathbb{SD}(X)$$

Don't forget the absolute values and squares!

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Properties of Variance #2

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The variance of a sum is affected by the dependence between the two random variables that are being added. Let's expand out the definition of Var(X + Y) to see what's going on.

Let $\mu_x = E[X], \mu_y = E[Y]$

$$
\begin{aligned}
Var(X+Y) &= E\big[(X+Y-E(X+Y))^2\big] \\
&= E\big[((X-\mu_x)+(Y-\mu_y))^2\big] \\
&= E\big[(X-\mu_x)^2 + 2(X-\mu_x)(Y-\mu_y) + (Y-\mu_y)^2\big] \\
&= E\big[(X-\mu_x)^2\big] + E\big[(Y-\mu_y)^2\big] + 2E\big[(X-\mu_x)(Y-\mu_y)\big] \\
&= Var(X) + Var(Y) + 2E\big[(X-E(X))(Y-E(Y))\big]
\end{aligned}
$$

By the linearity of expectation, and the substitution.

We see

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Addition rule for variance

If X and Y are **uncorrelated** (in particular, if they are **independent**), then

$$\mathbb{V}ar(X+Y) \;=\; \mathbb{V}ar(X) + \mathbb{V}ar(Y)$$

Therefore, under the same conditions,

$$\mathbb{SD}(X+Y) \;=\; \sqrt{\mathbb{V}ar(X) + \mathbb{V}ar(Y)} \;=\; \sqrt{(\mathbb{SD}(X))^2 + (\mathbb{SD}(Y))^2}$$

- Think of this as “Pythagorean theorem” for random variables.
- Uncorrelated random variables are like orthogonal vectors.

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